MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 5749W 1/

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Pentode, miniature, RF amplifier, remote cutoff

Outline --- 5-2 (EIA)
Base --- E7-1
Envelope --- T5-1/2
Cathode --- Coated unipotential
Base connections:

Pin No.	---	1	2	3	4	5	6	7
Element	---	g1	g3 sd	h	h	a	g2	k

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec1	Ec2	Ec3	Ehk	Rg1	Ic	Pp	Pg2	TE	Rk	Alt
Unit:	V	Vdc	Vdc	Vdc	Vdc	v	Meg	mAdc	W	W	°C	Ohms	ft
Maximum:	6.9	330	0,-50	165	0	100	1.0	1.0	3.3	0.7	165	---	(See note 1)
Minimum:	5.7	---	---	---	---	---	---	---	---	---	---	---	
TEST CONDITIONS:	6.3	250	0	100	(See note 2)	---	---	---	---	---	---	68	---

GENERAL:

Qualification - Required

Reliable tube

1/ Formerly tube type 5749

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1							
1256	Electrode current (1) (anode)		0.4	II	Ib	8.5	13.5	mAdc
1256	Electrode current (screen)		0.4	II	Ic2	---	5.6	mAdc
1266	Total grid current	Ec1 = -1.0 Vdc; Ec2 = 120 Vdc; Rg1 = 0.25 Meg (see note 3)	0.4	II	Ic1	0	-1.0	µAdc
1301	Heater current		0.4	II	If	275	325	mA
1306	Transconductance (1)		0.4	II	Sm	3600	5200	µmhos
1336	Heater-cathode leakage		0.4	II	Ihk	---	10	µAdc
1201	Short and discontinuity detection		0.4	II	---	---	---	---
	Quality conformance inspection, part 2							
1211	Insulation of electrodes		2.5	I	---	---	---	---
1031	High-frequency vibration	Rp = 2,000 ohms	6.5	Code E	Ep	---	300	mVac
2201	Noise and microphonics	Ef = 6.3 Vdc; Ebb = Ecc2 = 300 Vdc; Eca1 = 200 mVac; Rk = 200 ohms; Rp = 10,000 ohms; Rg2 = 60,000 ohms; Cg2 = 2 µF	2.5	I	---	---	---	---
1266	Grid emission	Ef = 7.5 V; Ec1 = -25 Vdc; Rg1 = 1.0 Meg; Rk = 0; Ck = 0 (see note 4)	2.5	I	Isc1	0	-1.0	µAdc
1306	Transconductance (2)	Ef = 5.7 V	2.5	I	ΔSm Ef	---	15	%
1306	Transconductance (3)	Ec1 = -20 Vdc; Rk = 0	2.5	I	Sm	5	100	µmhos
1331	Direct-interelectrode capacitance	No shield	6.5	Code E	Cg1p	---	0.0035	pF
					Cin	4.4	6.6	pF
					Cout	3.5	6.5	pF
1121	Base strain	See note 5	---	---	---	---	---	---
2126	Envelope strain		2.5	I	---	---	---	---
1041	Shock	450 G; Ehk = +100 Vdc; Ck = 0 (see note 6)	6.5	See note 7	---	---	---	---
1031	Vibration fatigue		6.5	See note 7	---	---	---	---

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
						MIN	MAX	
	Quality conformance inspection, part 2 -Continued							
---	Shock and vibration-fatigue test end points:							
1031	High-frequency vibration		---	---	Ep	---	450	mVac
1336	Heater-cathode leakage		---	---	Ihk	---	30	µAdc
1306	Transconductance (1)		---	---	Sm	3000	---	µmhos
1266	Total grid current		---	---	Ic1	0	-2.0	µAdc
1105	Permanence of marking		---	---	---	---	---	---
	Quality conformance inspection, part 3							
1506	Heater-cycling life	Ef = 7.5 V; Ehk = +135 Vdc; Ec1= Ec2 = Eb = 0			---	---	---	---
---	Heater-cycling life-test end point:							
1336	Heater-cathode leakage				Ihk	---	20	µAdc
1516	Stability life	Eb = 300 Vdc; Rg1 = 1.0 Meg; Ec2 = 150 Vdc; Ehk = +135 Vdc; Rk = 230 ohms; Ck = 0; TA = room			---	---	---	---
---	Stability life-test end point (2 and 20 hours):							
1306	Change in transconductance (1) of individual tubes				ΔSm t	---	10	%
1501	Intermittent life	TE = 165°C (min) (see note 8) Group E Eb = 300 Vdc; Rg1 = 1.0 Meg; Ec2 = 150 Vdc; Ehk = +135 Vdc; Rk = 230 ohms; Ck = 0; TA = room			---	---	---	---
---	Intermittent life-test end points (1,000 hours):							
---	Inoperatives				---	---	---	---
1266	Total grid current				Ic1	0	-1.0	µAdc
1301	Heater current				If	275	333	mA
1306	Change in transconductance (1) of individual tubes				ΔSm t	---	25	%
1306	Transconductance (2)				ΔSm Ef	---	20	%
1336	Heater-cathode leakage				Ihk	---	20	µAdc
1211	Insulation of electrodes				R	50	---	Meg

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NOTES:

1. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage.

2. Tie grid 3 to negative terminal of cathode resistor.

3. This test shall be performed at the conclusion of the holding period.

4. Prior to this test, tubes shall be preheated a minimum of 5 minutes operating at the conditions specified below. The 3-minute test shall not be permitted. Test at specified conditions within 3 seconds after preheating. Grid emission shall be the last test performed on the sample selected for the grid-emission test.

Ef	Ec1	Ec2	Ec3	Eb	Rk	Rg1
V	Vdc	Vdc	Vdc	Vdc	Ohms	Meg
7.5	0	100	0	250	68	1.0

5. Acceptance sampling procedure shall be in accordance with "Base-strain test, miniature, sampling (method 1121)", except that data covered in "Acceptance and rejection criteria" shall be modified as follows:

   a Accepted if not more than one defective for class "A", "B", or "C" defects, respectively (see method 1121), or if not more than a total of two defectives are found in the sample.

   b Rejected if two or more defectives for class "A", "B", or "C" defects, respectively, or if a total of three or more defectives are found in the sample.

6. A grid resistor of 0.1 megohm shall be added; however, this resistor shall not be used when a thyratron-type short indicator is employed.

7. This test shall be conducted on the initial lot and thereafter on a lot approximately every 12 months. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes. When one lot has passed, the 12-month rule shall apply. MIL-STD-105, sample size code letter E, shall apply.

8. Envelope temperature (TE) requirements, when measured in accordance with the temperature by conduction-band measurement (method 1226), will be satisfied if a tube having bogey Ib (±5 percent) under normal test conditions, is determined to operate at or above minimum specified temperature at any position in the life-test rack.

9. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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